Structure of permanganate ion

Permanganate ion has tetrahedral geometry in which the central Mn^{7+} is d^3s hybridised.

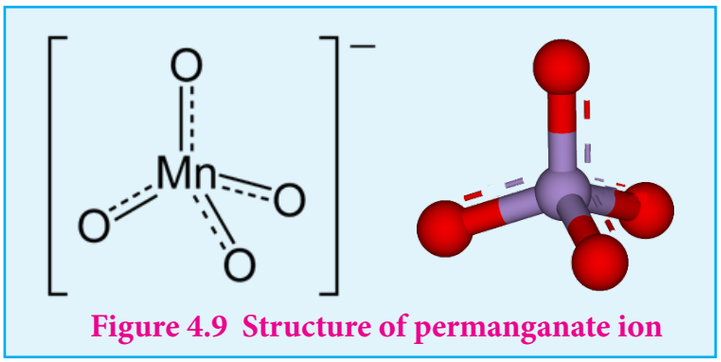


Figure 4.9 Structure of permanganate ion

Chemical properties:

1. Action of heat:

When heated, potassium permanganate decomposes to form potassium manganate and manganese dioxide.

$$2KMnO_4 \longrightarrow K_2MnO_4 + MnO_2 + O_2$$

2. Action of conc H_2SO_4

On treating with cold conc H_2SO_4, it decomposes to form manganese heptoxide, which subsequently decomposes explosively.

$$2KMnO_4 + 2H_2SO_4 \xrightarrow[\text{(cold)}]{} Mn_2O_7 + 2KHSO_4 + H_2O$$

$$2Mn_2O_7 \xrightarrow{\Delta} 4MnO_2 + 3O_2$$

But with hot conc H_2SO_4, potassium permanganate give $MnSO_4$

$$4KMnO_4 + 6H_2SO_4 \xrightarrow[\text{(hot)}]{} 4MnSO_4 + 2K_2SO_4 + 6H_2O + 5O_2$$

3. Oxidising property:

Potassium permanganate is a strong oxidising agent, its oxidising action differs in different reaction medium.

a) In neutral medium:

In neutral medium, it is reduced to MnO_2.

$$MnO_4^- + 2H_2O + 3e^- \longrightarrow MnO_2 + 4OH^-$$

(i) It oxidises H_2S to sulphur

$$2MnO_4^- + 3H_2S \longrightarrow 2MnO_2 + 3S + 2OH^- + 2H_2O$$

(ii) It oxidises thiosulphate into sulphate

$$8MnO_4^- + 3S_2O_3^{2-} + H_2O \longrightarrow 6SO_4^{2-} + 8MnO_2 + 2OH^-$$

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